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# The Impact of Ergonomic Rationalisation on the Efficiency and Productivity of the Production Process

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**Abstract:** This paper is aimed at understanding the possibility of applying ergonomics in the reorganisation of the work environment with the aim to improve working conditions and to increase the productivity of the examined workplace in an industrial company. Due to constant changes in markets, industrial companies are forced to seek new methods and paradigms for planning and managing innovations in order to ensure their competitiveness. An essential part of this process is the emphasis on improving production processes, where various methods with different focuses can be used. These methods not only optimise work processes, but also allow companies to minimise the resources needed for production and increase overall productivity. Another useful tool for industrial enterprises can be ergonomic rationalisation. The importance of ergonomics in improving employee working conditions and production process efficiency has been the subject of studies promoting various concepts. This study focuses in particular on examining the possibility of extending the outputs obtained by the REFA method to outputs obtained through ergonomic analysis. To achieve the objectives of the paper, the case study method was chosen, given that it was necessary to apply the REFA method in combination with ergonomic rationalisation in the specific conditions of the industrial company for the possibility of identifying bottlenecks in the production process from the point of view of its productivity, efficiency, and workforce involvement. Based on the results, it was possible to propose measures to increase the efficiency of the production process while respecting the principles of ergonomics. As part of the solution, the author team concluded that the findings obtained by combining both methods do not show significant differences, but rather complement each other and provide a broader view of the issue under study. At the same time, it can be stated that the solution cannot be considered definitive due to possible dynamic changes in the industrial environment (changes in the composition of the workforce and the scale of production and evolving technology, e.g., AI). The subject of future research will be to adapt the applied combination of methods so that it is universally applicable to any industrial sector, with minimal required adjustments to meet the specifics of individual industries.

**Keywords:** rationalisation; ergonomics; productivity; employees

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## 1. Introduction

Industrial enterprises are constantly seeking new methods and paradigms for planning and managing innovation to effectively serve new and existing markets with new and/or changed products and services. Part of this process is an emphasis on improving the production processes that form the basis of efficient business operations. Given the need

to keep pace with the current challenges of the times, it is essential to purposefully seek out bottlenecks and waste within the production process and, ideally, eliminate them. This can be achieved by implementing modern manufacturing technologies such as automation (Ajiga et al., 2024; Karumban et al., 2023), digitalisation (Lee et al., 2021; Pihir et al., 2018), and lean manufacturing principles (Varela et al., 2019; Lai et al., 2019). These methods not only optimise work processes, but also allow companies to minimise the resources needed for production and increase overall productivity. Rationalisation can also be another useful tool for industrial enterprises.

Rationalisation of production has its justification in the development of the company, and thus, improves its competitiveness in the market. These interventions in production offer the possibility of using new technologies which are beneficial both in terms of improving working conditions and in economic terms (Dostal & Sadilek, 2021).

Rationalisation can take different forms, depending on its focus and objective. In general, rationalisation can be defined as a set of state-of-the-art knowledge, methods, techniques, and organisation that are designed to ensure minimal wastage of effort (human effort) or material (cost reduction) in order to increase benefits (Rolander et al., 2013; Szombathyová & Krauszová, 2008).

The aim of the paper is to understand the potential of the comprehensive application of ergonomics in the reorganisation of the work environment. The purpose is to improve working conditions and to increase the productivity of the studied workplace in an industrial enterprise.

The research question guiding this study is as follows:

Question 1: Will the application of another method enrich the results obtained by applying the REFA method?

When formulating the research question, we assumed that by applying a combination of methods, more robust and comprehensive data could be obtained to influence the efficiency and productivity of the production process using ergonomic rationalisation.

## 2. Theoretical Background—Production Processes and Possibilities for Their Rationalisation to Increase Efficiency and Productivity

The classification of individual types of production processes according to several criteria allowed us to reveal some of their characteristics that can be generalised to other types of production processes. Starting from a rather general definition of a product, we will refer to these generalised manufacturing systems as production systems. Irrespective of the substantive nature of the product, it is always a complex of purposeful activities of organised groups of people, which is divided into operations and sub-processes. It is a given complex of operations (examinations at specialised workplaces) to be carried out in a certain order (Součková, 2017).

The manufacturing process represents the activities associated with the actual design of the product, the technology of its manufacture, the production process, assembly, testing, packaging, and shipping. In order to ensure that the entire production process is carried out correctly in a manufacturing enterprise, the entire production stage must be taken care of in a timely manner. This is where the technological changes to the starting materials are made. This is also where the components and the whole product are given the shape or composition specified by the designer in the technical documentation in order to achieve the desired level of quality and utility at the minimum production cost. From the point of view of business economics, the aim is to realise economically optimal production (Součková, 2017).

The connection between production processes and ergonomic rationalisation is key to optimising production efficiency, improving working conditions, and increasing employee

productivity (based on Brito et al., 2018). This approach combines the technical aspects of production with ergonomic design principles, ensuring that production processes are efficient, safe, and adapted to human abilities and constraints. Achieving greater efficiency through ergonomic rationalisation requires linking the scientific knowledge of human performance, job design, and work organisation.

In this case, rationalisation can be defined as the process of streamlining and improving the productivity and efficiency of activities in an organisation through the use of process adjustments, improved outputs, better planning and coordination, more efficient use of resources, the introduction of new technologies, and improved management and organisation of work (Vaara et al., 2006).

If the work activity is the subject, rationalisation may involve altered requirements for how leadership is exercised and rearrangements of work practices and work tasks, often at the cost of intensifying work through altered time exposure factors (Rolander et al., 2013).

The aim of rationalisation can be to create a better and more efficient working environment for employees and to improve the performance of the organisation as a whole.

When we need to make the involvement of the workforce more efficient, we can define rationalisation as the improvement of human activity, increasing its efficiency and economy, i.e., the totality of measures for the most efficient use of manpower and technology on the basis of the use of modern knowledge of science and new technologies. It is in this process that ergonomics can be used, and its application is changing with the growth of research and knowledge about ergonomics but also in the light of new human problems arising around the world (Koirala & Nepal, 2022).

The objectives and benefits that can be achieved through rationalisation are illustrated in Figure 1.

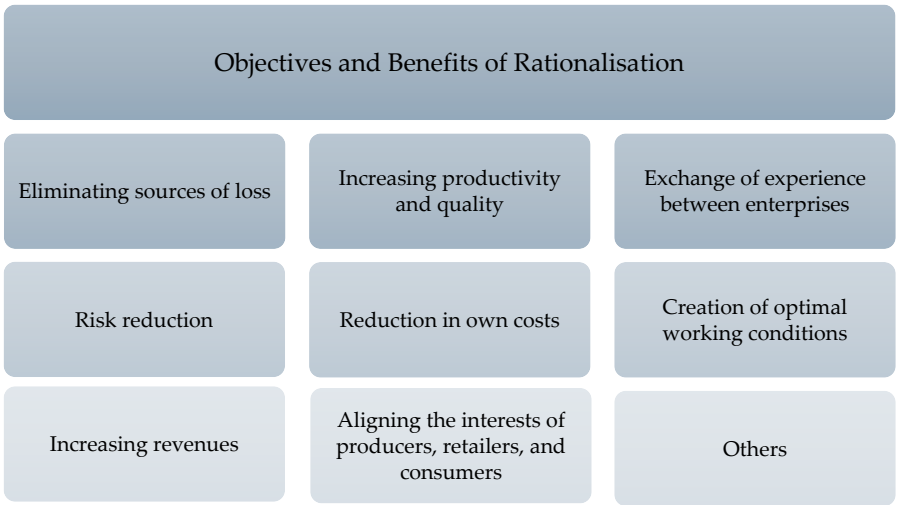


**Figure 1.** Objectives and benefits of rationalisation (own processing by Szombathyová & Krauszová, 2008).

Ergonomics in its focus can cover a wide range of topics related to the workforce. It studies the relationship between workers and jobs, which affects every part of the workplace. The subject of ergonomics can be the design of the workplace itself, the equipment of the product, the environment, and personnel policies, taking into account the biomechanical, physical, and psychological needs of employees. When all aspects are taken into account, it is possible to improve the efficiency and productivity of the work system while ensuring worker health, safety, and satisfaction (Koirala & Nepal, 2022; Onofrejová & Šebo, 2020).

Like ergonomics itself, perspectives on the working environment have evolved over time. In the work environment, the focus is not only on its functionality but also on its aesthetic appeal, as a good-looking space can motivate, inspire, and ultimately increase a person's work efficiency (Peteri, 2017).

However, for many companies, ergonomics can be seen as an external factor that is not part of their strategy, planning, and control cycles (Koirala & Nepal, 2022). The role of ergonomics is to create a comfortable working environment for the worker. On the other hand, with the advancement of technology, progressive technologies such as simulations, virtual reality, and augmented reality are offered in Industry 4.0. The advantage is the ability to create a virtual environment of any work process. Therefore, integrating Industry 4.0 technologies with ergonomics can increase workplace efficiency. Ergonomists can use these simulations to identify risks or sources of discomfort that may affect the actual worker (Mostafa, 2023).

The environment in which employees work has a huge impact on work efficiency as well as on comfort and safety. If the work environment is not designed correctly, the employees compensate for the deteriorated conditions, which in turn affects their health and comfort, and ultimately, affects their overall productivity and efficiency. In such a context, the workspace can also be seen as a place used to continuously look for room for improvement. There are a number of approaches that address the underlying determinants of the work environment. Villarouco et al. state ergonomics as supporting the following (Villarouco et al., 2012):

- Natural comfort—warmth, light, or suitable acoustics;
- The dimensions of the working environment in relation to the type of work to be performed;
- The materials used;
- Equipment and furniture and their distribution within the space;
- Safety;
- Sustainability.

Rationalisation of work is expected to increase efficiency, competitiveness, and sustainability in a dynamic business environment. Work rationalisation can be achieved through individual predetermined methods and procedures to be followed. For maximum efficiency in production, rationalisation measures such as improvement in the spatial layout, work organisation, work processes, work products, etc. are used. This is mostly based on the suggestions or objections of workers, preliminary economic calculations, etc. (Herman, 2012).

Within ergonomics, there are many tools and methods that are used to assess the work environment and ensure the safety and comfort of employees. A number of modern technologies (Table 1) in the form of simulation software with the ability to monitor a larger number of parameters can be used to collect the data needed to evaluate the interrelationships between the employee, the work environment, and the work tools. Although the perspective of the worker is extremely important, the use of these technologies objectifies the overly subjective nature of the assessment. Importantly, it is desirable to combine several of the above technologies for objective data collection and subsequent evaluation. From the collected data, a digital working environment can be modelled. We can incorporate a digital employee with specific physiological characteristics into this environment. A number of tests and simulations can then be performed, and the ergonomic suitability of the working environment can be monitored. The advantage is that there is no need to visit real work areas, and there is the possibility to simulate different work tasks (Berlin & Adams, 2017; Horváthová et al., 2018).

**Table 1.** Tabular representation of technology types used for data collection in ergonomic assessment (Horváthová et al., 2018; Middlesworth, 2024).

| Technology Type | Functionality | Basic Evaluated Parameters | | | | | | | | | Impact on the Work Activity Being Monitored<br>* None<br>* Partial | Methods and Ways of Evaluation |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Position of the Upper Limbs | Position of the Lower Limbs | Torso Position | Head and Neck Position | Use of Force | Frequency of Movements | Energy Expenditure | Subjective Assessment of the Burden | Workplace Layout | | |
| Wearable sensor systems | Sensors for measuring physiological functions | | | | | * | * | * | | | * Discomfort caused by wearing sensors | Legislative, the Ruffler method, Index method |
| Motion capture | Sensing human position and movement | * | * | * | * | | * | | | | * Sensing without markers<br>* Sensing with markers | Legislative, OWAS, RULA, REBA |
| CERRA | Augmented reality workplace screening | * | | * | * | | * | | | * | * Observer presence at the workplace | Legislative, OWAS |
| Simulation software | Modelling and simulation of work activities and the workplace | * | * | * | * | * | * | | | * | * | Legislative, OWAS, RULA, REBA, NIOSH |
| Mobility applications | Questionnaires and checklists | | | | | | | | * | * | * | Legislative, Nordic questionnaire |

Efficient production development is a key aspect of a successful business. Rationalisation, as a tool for optimising production, involves rational decision making, the promotion of worker initiative, and professional rationalisation activities. Rational action consists of the optimal use and creation of the individual parts of the system that is subject to rationalisation. It is crucial for successful development that the objectives of rationalisation and change are properly understood and that the conditions for their realisation are created. The economic essence of rationalisation efforts consists of achieving higher performance and efficiency through the knowledge and application of modern systems, concepts, and working methods.

## 3. Materials and Methods

To increase the efficiency of the production process, emphasis is now being placed on optimising and rationalising production schedules. In addition to time analysis, it is also necessary to take into account the human factor, which can be limiting in terms of the success of the solution. In practice, several methods can be used to analyse production processes to make processes more efficient, reduce waste or improve quality, or ensure the smooth operation of the company. Time and motion study (TMS), Value Stream Mapping (VSM), Six Sigma Analysis, DMAIC, Work Sampling, Failure Mode and Effects Analysis (FMEA), Statistical Process Control (SPC), the REFA method, lean manufacturing, and Bottleneck Analysis are the most commonly used methods for the analysis of manufacturing processes, and their results are mainly translated into indicators such as Overall Equipment Effectiveness (OEE), Process Capability Analysis (Cp/Cpk), and others.

The selection of the appropriate method for analysing the production process depends on the desired objectives and the industry in which it will be used. Often, it is necessary to use a combination of several methods to take a holistic view of the process and ensure continuous improvement.

Due to the nature of manufacturing and the industry in which the case study was addressed, the REFA method, which is based on detailed time measurement and includes analysis to identify inefficient movements or steps, was used for the analysis. It focuses on structuring work tasks to increase productivity.

The REFA method (Reichsausschuß für Arbeitszeitermittlung) (Bures & Pivodova, 2015; Vlčeková, 2024) is a method of direct measurement of work with the ability to assess with what intensity and efficiency the work is performed. It is applicable to both cyclical and non-cyclical work. The REFA method (REFA, 2003) details the steps involved in conducting time studies. It is widely used within industrial engineering to improve the efficiency of manufacturing processes (based on Čolović et al., 2024). It is part of the broader field of time and motion studies and involves systematic analysis to optimise work flows, reduce waste, and increase productivity. The method relies heavily on the detailed recording and observation of work processes and is useful for designing better work environments and organising tasks more efficiently. In particular, it includes an unambiguous system of nomenclature and division of time, performance rating, and special recording forms.

We have therefore used the REFA method to analyse work processes in detail and identify areas where efficiency and productivity can be improved. Comparison of the outputs with legislation will ensure compliance with standards and regulations relating to workers' working conditions, complementing the results from the CERAA application.

For the purpose of the analysis, the work of all work activities at each workstation within production lines A and B of the production plant was measured in the months from July to September in 2023. The parent company of the company examined in the study is located in Germany, where the REFA method is standardly used to study work activities. The company thus has a database of time snapshots of the production lines,

which made it possible to choose a quantitative approach. Measurements for the needs of the case study were carried out in the branch company under study. The following methodological procedure was applied when implementing the REFA method. As part of the preparation of the analysis, its goal was chosen, in this case, reducing working time. Work activities within the production process were selected for the needs of the analysis. Subsequently, all relevant data on the process (time, costs, tools, and methods) were collected. As part of the work process analysis, the process was mapped to obtain a complete overview of all its steps. In order to identify tasks and operations, the process was divided into smaller tasks and operations performed by the worker. Then, the times of individual operations were measured using stopwatches. Data collection was carried out using a Casio stopwatch HS-80TW-IEF (Casio Computer Company, Limited, Sibuji, Tokio, Japan) and a pre-prepared REFA time frame in the company's production hall during the relevant production operations on the line. Individual activities on the production line were measured between 25 and 30 times, depending on the type of activity measured. As long as an operation performed on one piece was measured, it was measured 30 times, and as long as different activities within one operation were measured on several modules at the same time, the measurement was repeated 25 times. The analysis of working conditions was processed using the REFA form, where information about the work task, work time, and worker were recorded (with respect for GDPR); the work process was also summarised, the influences acting during the measurement were recorded, and notes with the form coding were made for traceability. The form included a time snapshot intended for serial production with regular repetition, where the sections of the process were identified, numbered, and defined, and measurement points were determined. It was also necessary to determine the relative quantity (number of modules on the carrier, 1 piece when welding) and, finally, the level of required performance and times. For the purpose of analysis, all 24 workers working at each position on the production line were imaged. The age range of the female workers was 35–55 years, laterality was right-handed in all of them, and work exposure ranged from 0 to 12 years. Furthermore, 66% of the female workers performed work sitting, and 34% of the female workers stood while working. None of the female workers were trained in the occupation, but 23 female workers had been properly trained for the job. Thus, it was assumed that they are well aware of the production processes and have sufficient experience of them as a result of their occupational exposure. One female worker was part of the training programme for one week, and her knowledge and skills required to work on the imaging operation were assessed at 85%; thus, she did not perform all the operations to full capacity. Her work performance was slightly limited, the duration of individual operations was longer, and the cost required to perform them was higher. Adjusting time and cost estimates in light of the worker's actual performance will help to obtain a realistic picture of processes and costs. For a comprehensive view of worker performance, measurements were taken at different times of the day (at the beginning of the shift, before and after the lunch break, and at the end of the shift). By averaging the performance curve from the collected data, it was possible to refine the measurement results with respect to the whole working day. The next step was to improve the work process by identifying inefficient and redundant movements and actions. The production process was subsequently modified by redesigning the work process by introducing new technologies and changing working conditions. In the phase of determining the standard time, it was calculated for the individual operations under study based on measurement and analysis, taking into account productivity standards. As part of the implementation of the improvements, employees were retrained in new work procedures and new technologies. Control and verification of the correctness of the solution were carried out by simulation using the Witness software (v.12).

In addition to the measurement of work activities, an analysis of the risk factors of the working conditions and their projection to individual body parts was carried out at the workplace using data collection through the Nordic Questionnaire (Kuorinka et al., 1987). All (100%) of the female workers participated in the analysis. The findings were supported by screening of the ergonomic conditions at the analysed workplaces through the use of the CERAA application. In this way, the spatial design of the production line workplaces was analysed; specifically, the reach distances were measured. The analysis was carried out on all four operations, assessing the spatial conditions for female workers, including an assessment of the work area when working both sitting and standing.

## 4. Results

The following part of the paper includes the results of the analysis of the production process by the REFA method, through which time was measured in detail and inefficient movements or steps were identified. Based on this, it was possible to structure work tasks to increase productivity. To comprehensively improve working conditions and increase overall productivity on the production line, the findings were complemented by a detailed ergonomic analysis of the production line operation with emphasis on identifying areas for improvement. Consequently, it was possible to propose measures to increase the efficiency of the production process while respecting the principles of ergonomics.

### *4.1. Results of the REFA Manufacturing Process Assessment*

The next section of the paper covers the outputs of the analysis using the REFA method. The aim was to identify bottlenecks on sections of the line in terms of differences in times between operations and to rationalise an inefficient process within the production cycle in the form of multiple carrier handling that does not add value to the final product.

The reason for using the REFA method to improve the efficiency of the production process and productivity in the company is its standard use in the parent company in Germany. Within the company database, there is a database of REFA time frames for production lines; therefore, a quantitative approach using the method was chosen. For the outputs within the case study, the measurements were performed in the present. The data collection was carried out using a stopwatch equipped with a precision measuring mechanism, which enables accurate time recording, in a pre-prepared REFA time frame in the production hall of the company during the production operations in question on the line.

The work measurements were taken on production lines 1 and 2 for all the activities taking place at the workstations. The flow of material starts on the right at the assembly area, where three workers (or more as needed) sit and assemble the torso. The torsion chassis are then moved along gravity conveyor 1 to the furnace workstation. The furnace workstation is not the subject of the analysis as the soldering furnace program cannot be shortened or influenced. The soldering furnaces are operated by a single worker, and the programme takes 60 min, after which the worker pulls out and puts the module stack on gravity conveyor 2. From there, a worker (green colour in Figure 2) takes the spool and removes the semi-finished product from the process and assembly torsion—line section B. The semi-finished products are transferred on a carrier to two workers on line section C (yellow), who place each piece near the workstation on line section D (orange). During production on line 2, a worker carrying out activities for line section C (yellow) carries the semi-finished products in a carrier to the worker at the ultrasonic welder (orange, worker on the left).

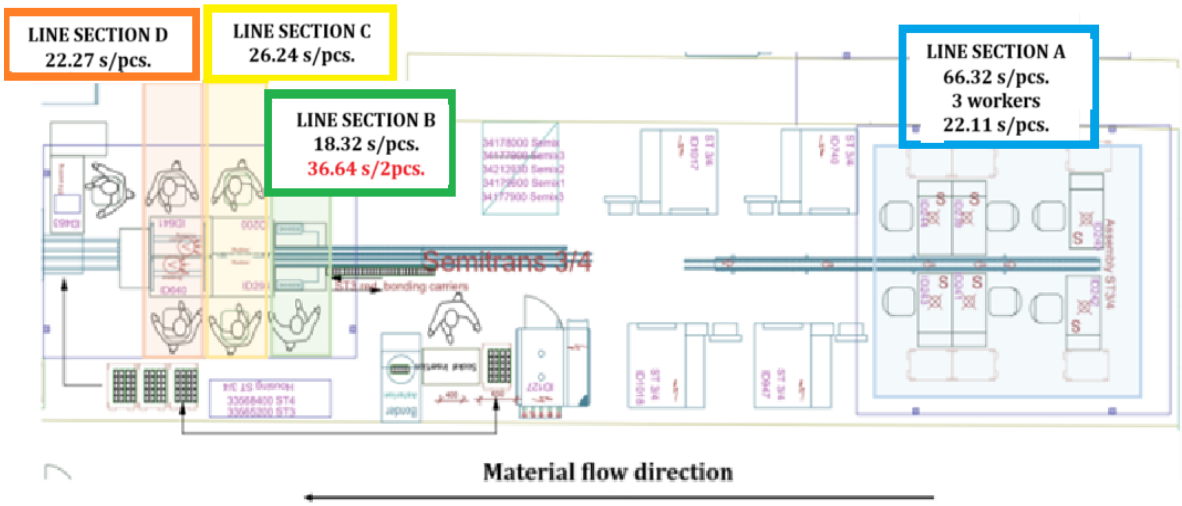


**Figure 2.** Time snapshot results for ST3 in the workplace layout (own processing, 2023).

From the individual measurements, using a stopwatch, the individual time $t_i$ in a predetermined number was detected, from which the advancement time F was subsequently calculated. It is an internal company rule that the value of ε must be less than or equal to 2.5% when processing the time frames. This rule was followed during data collection.

After filling the form with the baseline and measured data, the evaluation is calculated in the right part of the spreadsheet according to the progression section in the time frame sheets.

The first step represents the sums of the assessed performance level L at the progression segment, where L = 100 (percent of worker performance) and n represents the number of measurements.

$$\sum L_n = 100 + 100 + 100 + \cdots = 100 * 25 = 2500. \quad (1)$$

In the case of a worker with 85% performance, L = 85. This also applies to the other calculations. Similarly, the sum of the actual individual times $t_i$ of the progression sections must be calculated, where the individual measured times are summed, as shown in the example from time frame ST3-R-2 for the operation "Taking the splint off the belt".

$$\sum t_{in} = 4.28 + 4.58 + 7.52 + 6.94 + 3.93 + 6.56 + 3.42 + 4.96 + 6.92 + 7.26 + 10.3 + 9.35 + 6.84 + 3.82 + 5.75 + 5.98 + 5.77 + 4.99 + 12.49 + 6.36 + 5.08 + 4.31 + 5.97 + 6.81 + 6.77 = 156.94, \quad (2)$$

The second step is the determination of the mean power level L according to the corresponding calculation, where ∑L represents the sum of the considered power level at the advancing section and n in the calculation represents the number of the considered power level. In all time frames, the same calculation procedure was used, shown for the line section B on production line 1, registration number ST3-R-2:

$$\overline{L} = \frac{\sum L}{n}; \ \overline{L} = \frac{2500}{25} 100 \quad (3)$$

The mean value of the actual individual time $\overline{t_i}$, where $\sum \overline{t_i}$ represents the sum of the actual individual times of the progression legs, and n in the calculation represents the number of summed actual individual times, was then determined. In all time frames, the same calculation procedure was used, shown for line section A on production line 1, registration number ST3-R-2:

$$\overline{t_i} = \frac{\sum t_i}{n}; \ \overline{t_i} = \frac{156.94}{25} = 6.3 \quad (4)$$

The required time t, which was given in seconds in the time slide sheets, was calculated using the mean power level and the mean value of the actual individual time $\overline{L}a\overline{t_i}$ according to the following formula:

$$t = \frac{\overline{L}}{100} * \overline{t_i} \tag{5}$$

For the ST3-R-2 time frame, the calculation of the required time t was adjusted due to the fact that there were 5 modules on the shield, and the calculations were recalculated on a per-piece basis:

$$t = \frac{\overline{L}}{100} * \frac{\overline{t_i}}{5} = \frac{100 * 6.8}{100 * 5} = 1.4 \tag{6}$$

Twenty-four measured time snapshots were evaluated in the same manner as for the ST3-R-2 snapshot, and the resulting average times of the monitored workers are recorded in Table 2.

**Table 2.** Resulting average times from time snapshots (own processing, 2023).

| Operation | Registration No. | Time (s) | Average (s) | Registration No. | Time (s) | Average (s) |
|---|---|---|---|---|---|---|
| Line section A | ST3-M-1<br>ST3-M-2<br>ST3-M-3 | 63.14<br>67.57<br>68.26 | 66.32 | ST4-M-1<br>ST4-M-2<br>ST4-M-3 | 75.32<br>78.26<br>75.37 | 76.32 |
| Line section B | ST3-R-1<br>ST3-R-2<br>ST3-R-3 | 18.00<br>17.45<br>19.50 | 18.32 | ST4-R-1<br>ST4-R-2<br>ST4-R-3 | 16.92<br>18.09<br>20.28 | 18.43 |
| Line section C | ST3-I-1<br>ST3-I-2<br>ST3-I-3 | 24.95<br>27.18<br>27.24 | 26.46 | ST4-I-1<br>ST4-I-2<br>ST4-I-3 | 26.06<br>23.68<br>24.87 | 24.87 |
| Line section D | ST3-Z-1<br>ST3-Z-2<br>ST3-Z-3 | 21.61<br>23.95<br>21.25 | 22.27 | ST4-Z-1<br>ST4-Z-2<br>ST4-Z-3 | 25.19<br>24.92<br>24.92 | 25.01 |

The times are charted on the line layout for the production of ST3 modules in Figure 2 and for the production of ST4 modules in Figure 3. The bottleneck, disassembly on line section B, is marked in the aforementioned figures, and this is because one worker disassembles and delivers semi-finished products for two lines of other operations, and hence, the doubled operation time is the longest compared with the others. Therefore, it may happen that the workers performing the operation on the line section C sometimes have to wait, which is considered wastage. Although the operation on line section A takes the longest time, it is supported by three workers, which means that one piece comes out from them every 22.11 s.

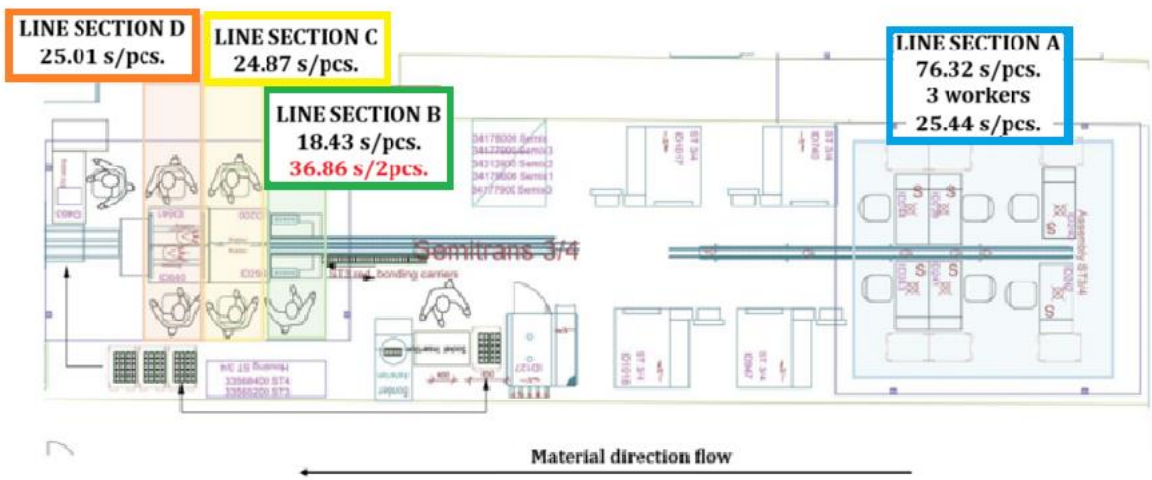


**Figure 3.** Time snapshot results for ST4 in workplace layout (own processing, 2023).

The bottleneck represents line section B. By evaluating the time frames, it was found that it takes a worker up to 3.26 s per piece to reload a carrier. The given time was calculated as the sum of the average times per piece obtained from the time snapshots from all the workers on the operation on line section C, specifically from the following activities:

- Taking the carrier off the table: (0.8 + 1.1 + 1.4 + 1.2 + (1.3 * 0.85))/6 workers = 1.1 s/pc,
- Unloading pieces for the next operation + carrier return: (2.2 + 1.9 + 2.4 + 2.2 + 2.1 + (2.55 * 0.85))/6 workers =2.16 s/pc.

Carrier reloading activity has potential for improvement. In one eight-hour shift, workers produce 1473 ST3s and 1465 ST4s, which means that in a three-shift operation, up to 4419 ST3s and 4395 ST4s are produced in one day. In a day, the workers of section C handle the carrier in a cumulative time of 4 h, i.e., 80.03 min in one shift for the production of ST3 and 79.60 min in one shift for the production of ST4.

In the case study, the aim was to rationalise an inefficient process within the production cycle not only in terms of time, but also in terms of workforce participation and comfort. If only the REFA method had been used to achieve the rationalisation objective, the inefficient process would have been addressed solely in terms of technical parameters and without taking into account the needs and constraints of the workforce. The rationalisation could be less efficient precisely because of the failure to take into account the workplace constraints on the workforce that limit the solution. This can be avoided by supplementing the analysis with an ergonomic assessment of the conditions on production lines A and B.

### *4.2. Ergonomic Assessment of the Production Line*

For a comprehensive assessment of the working environment with the aim of improving working conditions and increasing overall productivity on the production line, an ergonomic analysis was also performed using multiple methods.

When assessing workplace comfort from the perspective of spatial design and the impact of the risk factors in the work environment from the subjective perspective of the female workers, it was found that the highest incidence of body-related problems is in the neck and upper back (Figure 4), where 24% of the female workers were forced to see a doctor due to the aforementioned difficulties, which indicates their intensity.

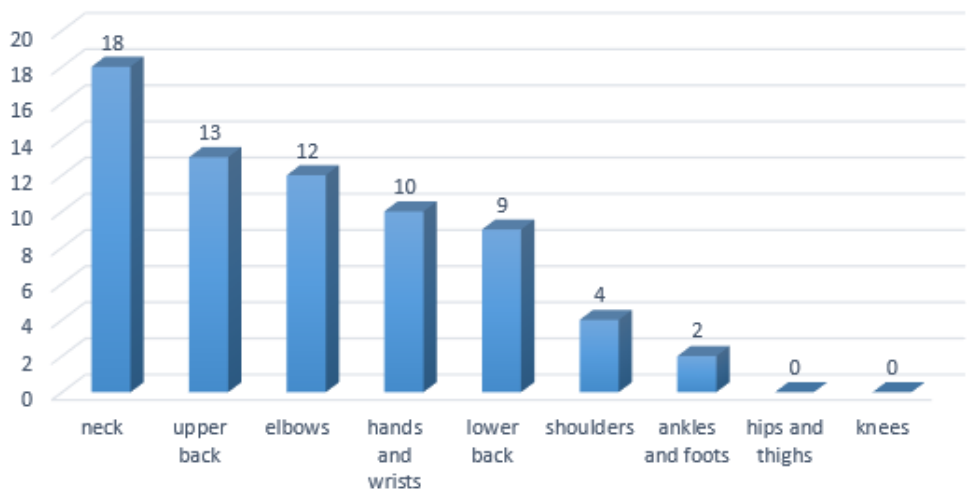


**Figure 4.** NQ—The pain and tingling in a specific part of the body (own processing, 2024).

To identify potential risks in the work environment, the female workers themselves subjectively assessed the work environment factors. The assessment score was in the range of 0–10, where factors rated above 9 points were considered bothersome. The results show that the greatest strain is caused by 3 factors, namely, a fixed working position without the

possibility of individual adjustment, the frequency of performing the work operation, and the handling of small objects.

The objectification of the identified shortcomings in workplaces and of the working environment conditions was ensured by screening using the CERAA application and confrontation with legislative documents. The non-compliance of the workplace solution with the legislation is shown in Table 3. The workplace screening itself supports the findings from the questionnaire survey, because as a result of the identified spatial deficiencies, discomfort is projected onto the above-mentioned body parts of the female workers.

**Table 3.** Evaluation of measurements of the sitting workstation dimensions (own processing, 2024).

| Working Plane Dimension—Sitting | CERAA (mm) | | Line Section A (mm) | Line Section B (mm) | Line Section C (mm) | Line Section D (mm) |
|---|---|---|---|---|---|---|
| | Optimal | Maximal | | | | |
| Height | 590 | 750 | 730 | 750 | 750 | 750 |
| Width | 685 | 1100 | 1000 | 760 | 1000 | 1000 |
| Depth | 315 | 500 | 480 | **530** | **540** | **510** |

At the workplace on line section B, it is possible to perform work while standing also; therefore, the corresponding dimensions of the workplace were measured for standing work as well. The results are given in Table 4.

**Table 4.** Evaluation of measurements of the dimensions of the workplace on line section B in a standing position (own processing, 2024).

| Working Plane Dimension—Standing | Non-Adjustable Working Height | | Space for Legs | | | |
|---|---|---|---|---|---|---|
| | | | Height | | Depth | |
| | CERAA (mm) | Real Size (mm) | CERAA (mm) | Real Size (mm) | CERAA (mm) | Real Size (mm) |
| Height | 900 | 970 | 226 | 250 | 181 | 226 |

By evaluating the data obtained from employees using the Nordic Questionnaire, the risk factors in the workplace appear to be the monotony of the work with long-term, excessive unilateral strain on muscle groups in the same working positions. The result is an increase in discomfort and pain, especially in the neck, upper back, and elbows. The cause was identified, also using the CERAA application, in the form of an unsuitable working plane height and unsuitable reach distances. Other dimensions of the workplaces are in accordance with legislative requirements. Nevertheless, the shortcomings in the area of reach distances signal the need to adjust working conditions to improve the ergonomics and worker safety. They also complement the findings that were identified through the application of the REFA method. It found:

- Existing differences in times between individual operations indicate a bottleneck on line section B. Because the time of a given operation is on a selected line section, the female worker takes the preparation for two branches of section C and section D, which can cause delays in the aforementioned operations, leading to a waste of time.
- An inefficient process within the production cycle exists; specifically, the multiple manipulations of the carriers by female workers translate to a cumulative time of 4 h per day, not including the manipulation of products associated with the use of the carriers, while the given activity does not add value to the resulting product.

Based on the above-mentioned outputs from the implemented analyses, it can be stated that the ergonomic analysis (implemented by combining the subjective views of the workers, objectified by applying the CERAA application) in combination with the REFA method made it possible to obtain more comprehensive data to influence the efficiency of

the production process. The assumption set out in the research question was confirmed by the application of the above-mentioned methods.

The findings revealed were also confirmed through a simulation of one work shift in the Witness Horizon simulation system. The identified outputs were also confirmed by observation and comparison with the applicable legislation in the area in question. Based on these data, it is possible to design a more efficient production process and rationalise the production line, not only from the perspective of its productivity but also the comfort of the human workforce, specifically:

- A conveyor system and electric conveyor would make it easier for female workers to handle modules, or even eliminate the handling completely in some steps, which will result in a reduction in the time per piece in operations on line sections B and C (Figure 5). We can eliminate the waste identified by the REFA method.

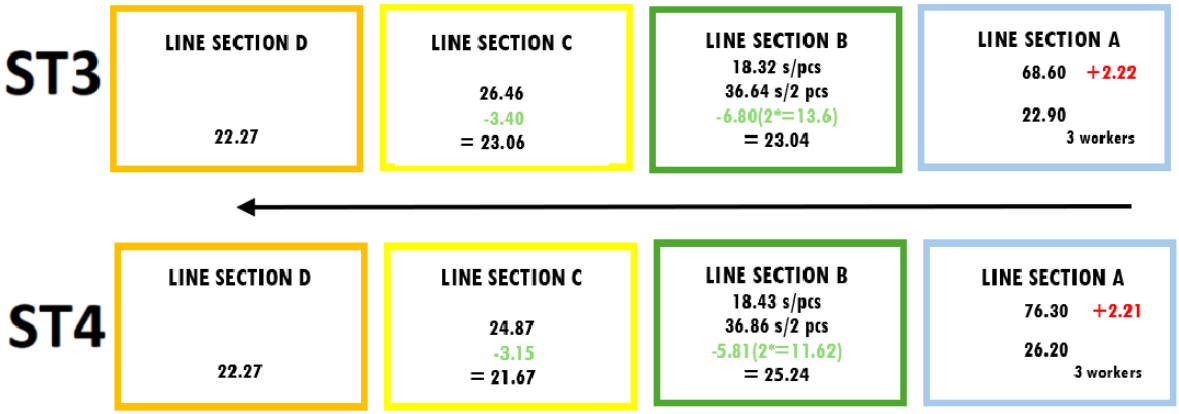


**Figure 5.** The conveyor system design summary (own processing, 2023).

By implementing the proposals, due to the bottleneck, 870 more pieces of product ST3 and 674 more pieces of product ST4 could be produced on the work line during one shift (Table 5).

**Table 5.** The comparison of the manufactured pieces on the line after modifications based on the results of the REFA method (own processing, 2024).

| | Product ST3 | | Product ST4 | |
|---|---|---|---|---|
| | Current State | After Adjustment | Current State | After Adjustment |
| Shift time without break (s) | 27,000 | 27,000 | 27,000 | 27,000 |
| Bottleneck (s) | 18.32 | 11.52 | 18.43 | 12.62 |
| Number of pieces produced per shift (pcs) | 1473 | 2343 | 1465 | 2139 |

- Ergonomic workplace equipment—proposals are aimed at improving the working environment and conditions for employees in the form of adjusting the design of the work desk (cutouts in the desk to reduce the depth of the working plane, elbow pads), adding local lighting and anti-fatigue ESD mats, and introducing rotation of the workers at exposed workplaces. The impact of an ergonomic solution on financial indicators may be indirect, but the impact on the overall efficiency and long-term sustainability of the company is significant. Ergonomic adjustments can increase the well-being and health of employees, which can have a positive impact on their work performance and satisfaction, thereby contributing to a comprehensive assessment of the work environment.

In the industrial company where the analysis was performed, it was decided to eliminate the identified bottleneck on line section B, which is caused by excessive walking when handling the semi-finished product, and also to divide the production batch into two branches of the operation. The bottleneck affected other operations on the production line. The waste in the form of redundant processes was found in the use of carriers that are used only between operations on the B and C line sections. The carriers took up space in the workplace, piled up on it, and handling them increased the ergonomic risks. To eliminate the identified shortcomings, an automatic conveyor system was designed to ensure transport between individual sections of the production line. Another proposal that will help rationalise the workplace and complement the conveyor system is the design of an electric conveyor that serves to move modules from line section A to line section B. The bottleneck identified on line section B, where it took the female worker up to 3.26 s per piece to remove and return a rail to the conveyor because it was far away from her, caused waste due to unnecessary processes, namely transfers and errors that occur when rails collide with each other. By combining it with the first proposal, a suitable workplace can be created for the worker on line section B, where she will have a rail with modules ready within reach as well as a place to return the rail.

## 5. Discussion

Published outputs declare numerous examples of the use of individual analytical methods, which are aimed at rationalising the production process either as a whole or in its parts. The very importance of rationalising production processes based on time analysis and ergonomics in improving the efficiency of the production process has been the subject of the various studies. When rationalising production processes, authors focus on automation of the production process (Ajiga et al., 2024), digitalisation (Lee et al., 2021), or lean production (Varela et al., 2019) in order to analyse the work process in detail and identify areas where efficiency and productivity can be improved. Ergonomic rationalisation deals with improving the work environment (e.g., lighting—Dupláková et al., 2019, 2022) and work processes (Novek & Bandurová, 2001) with the aim of increasing the efficiency, comfort, and safety of workers (Colim et al., 2021).

From the specific outputs within the case study, by applying the selected methods and implementing the proposed changes, the times between individual operations were shortened, the inefficient production process was rationalised, and the productivity of the work on the production line was increased while simultaneously adjusting the workplaces to eliminate the identified shortcomings in terms of ergonomics.

Bottlenecks identified in REFA can be evaluated very quickly through short-term indicators such as Cycle Time, which measures the time it takes to complete one unit of a product or process; Overall Equipment Effectiveness, which combines availability, performance, and quality to measure the efficiency of production equipment; Downtime Rate, which assesses the amount of time equipment is out of service; Scrap Rate, which tracks the number of non-conforming products; Throughput, the number of products produced in a given time; and Capacity Utilisation, which assesses the extent to which available resources (machinery, labour) are being utilised. The use of these indicators makes it possible to quickly identify problem areas and subsequently propose measures to eliminate them, thus improving the fluidity and efficiency of the production process.

The evaluation of the effectiveness of the proposed measures from the perspective of the benefit to the workforce cannot be assessed based on short-term indicators, because the effect will become apparent only after a certain period of time. We usually use a one-year evaluation period, when we can see whether the solution has brought positive impacts. For this reason, it seems appropriate to evaluate the social efficiency of an investment that

emphasises the human capital of a company. From a company perspective, social efficiency can be understood not as profit maximisation, but as creating added social value and social inclusion of the employees and reinvesting profits in the development of the main social goals of the company. The social responsibility encompasses an effective and responsible approach to the components of investments made for society, employee relations, creativity, and workplaces' sustainability. However, the effectiveness of investments in the field of ergonomics cannot be directly quantified using the classic arsenal of investment evaluation methods for several reasons:

- The effect of the investment does not appear immediately after the investment is made,
- The effect may manifest itself differently in each operation,
- The effect is almost always indirect, manifesting itself in areas such as reducing costs from illness (absence from work), reducing losses from accidents, reducing losses from employee turnover, and improving the work environment and work culture.

The assessment of the social efficiency of the expenditure of investment funds with respect to the useful result in a certain period of time (Michník, 1995) can be defined from the basic relationship:

$$E_{SOC} = u \,.\, e$$

where:

u—social effect of the investment,

e—effect per unit of the investment effect.

The social impact of an investment is usually defined as the area within which the measures taken are to be reflected in the form of an improvement in the identified/measured condition. Determining the unit of effect is no longer so clear-cut; it can be a specific (single) work position or a work cell to which the improvement is applied. Given that all of the above areas that fall into the area of social efficiency are quantified ex post on a quarterly, semi-annual, or annual basis, the effect can be measured only after they have been quantified, i.e., efficiency can be measured and quantified only after the final quantification of morbidity, injury, and fluctuation indicators. In addition to the fact that all indicators reduce the costs associated with production, it can be assumed that when "work does not hurt a person", i.e., does not cause any harm, the effect will also occur in the form of a reduced defective products rate and in the form of higher labour productivity.

## 6. Conclusions

Effective production development currently forms the basis of successful business. Rationalisation, as one of the tools for making production more efficient, means rational action, using workers' initiative and professionally performed rationalisation activities. Rational action is based on the optimal use and creation of individual components of the system that is the object of rationalisation. For its development, it is crucial that rationalisation plans and changes are correctly understood and that conditions and prerequisites for their implementation are created. The economic essence of the rationalisation efforts therefore lies in achieving higher performance and higher efficiency and presupposes the knowledge and application of newer and more modern systems, concepts, and methods of work (Szombathyová & Krauszová, 2008).

Compared with new technological trends such as Industry 4.0, the REFA method still has a relevant role in maintaining proportionality between human effort and benefit. It also suggests that, given the ever-evolving technology, the REFA method will remain relevant in the future and can be an effective tool for improving efficiency and optimising production processes in current industry, but it will likely require further development to adequately respond to new technological challenges (Ahrens, 2018). Between the two

major development lines in industry in recent decades, automation and the development of the methods to improve production efficiency, the REFA method falls into the latter group, as it focuses on improving production efficiency and optimising costs (Ahrens, 2018).

The current work reality reflects the growing use of artificial intelligence in the design of hybrid work systems. The REFA method and the periodic AI system form the basis for the analysis and optimisation of production processes, with an emphasis on supporting, not replacing, the human workforce. AI parts acquire diverse data, such as assembly information in the work environment, which they then use to generate information, situations, or states, such as selecting a workflow based on the acquired component data. They later react to the acquired and evaluated data, such as by visualising the next work steps. The combination of the REFA method and artificial intelligence allows for a structured view of the use of AI in work systems, highlighting its benefits in ergonomics and digital assistance systems. This synergy strengthens efficiency and innovation in the work processes (Pietschmann et al., 2022). The REFA method identified bottlenecks in the production process, which were also confirmed from the employees' perspective by performing an ergonomic analysis, which will allow for a more comprehensive solution that takes into account the needs and limitations of employees, thus avoiding tunnel vision when solving the problem. In future research, it will be possible to consider incorporating artificial intelligence into the evaluation of collected data for the purposes of applying the REFA method; however, in the part dealing with the ergonomic rationalisation of the working environment, research will be needed that would eliminate errors from excessive use of AI to assess the participation of human labour in the production process, given that we believe that AI is not yet at a level that could guarantee the relevance of the data.

Within the research limitations, our research may have been influenced by the gender imbalance of the sample of respondents as well as its individual variability. It is important to note that the gender imbalance of the sample may affect the generalisation of the findings to the wider working population. Regarding data analysis, everything was carried out in accordance with the requirements for this type of data collection and evaluation. We do not assume the occurrence of any errors in the analysis and processing of data. The above solution cannot be considered definitive due to the possible changing composition of the workforce, the scope of production, and evolving technology. Therefore, it is necessary to monitor development trends and correct the solution through future research. The subject of the research following the above case study is to adapt the applied combination of methods so that it is universally applicable to any industry with minimal required adjustments to meet the specifics of individual industries.

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